

No Uniform Interpolation for Linear Logic

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Abstract

The literature on uniform interpolation in linear logic is very scarce: the sole known result is that the multiplicative-additive fragment of linear logic has the uniform interpolation property, and so does its intuitionistic and affine variants [1]. We exhibit a simple formula of the unit-free multiplicative-exponential fragment of linear logic that cannot have any uniform interpolant in propositional linear logic. This example shows not only that propositional linear logic does not have the uniform interpolation property, but also that its intuitionistic and affine variants neither do.

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1 Introduction

Looking at known results for interpolation in linear logic and its fragments, a reader can find a modicum of work, chiefly on Craig’s and Lyndon’s interpolation and with very little on uniform interpolation. *Craig’s interpolation* states that if $A \vdash B$ is provable, then there exists a formula C built only from variables occurring in both A and B and such that both $A \vdash C$ and $C \vdash B$ are provable. *Lyndon’s interpolation* is a strengthening of this property taking account the polarity of variables. These two properties have been proved for first-order linear logic and studied in several fragments of this logic. Notably, Roorda [19] proved and disproved Lyndon’s interpolation in various fragments of propositional linear logic; Brotherston and Rajeev [2] used display logic to obtain Craig’s interpolation for the multiplicative and multiplicative-additive fragments of linear logic, as well as for classical logic (they then formalised these results working with Dawson [5]); Strassburger [22] provides in the calculus of structures for multiplicative-exponential linear logic a result corresponding to Craig’s interpolation; a proof of Craig’s and Lyndon’s interpolation for first-order linear logic, along with links to cut-elimination, has been developed by Saurin [20], who then considered with Fiorillo and Osorio-Valencia [7] this question in the setting of the proof-net syntax of multiplicative linear logic. Let us also mention that Fussner and Santschi [8] researched Craig’s interpolation in extensions of classical and intuitionistic propositional linear logic using algebraic techniques.

The literature is even more scarce on uniform interpolation in linear logic. *Uniform interpolation*, introduced by Pitts [18] in the setting of intuitionistic logic, is a stronger version of Craig’s interpolation where the interpolant depends only on one of the two formulas under consideration. More precisely, a uniform interpolant for a formula A , with respect to a variable X , is a formula C such that:

- the variables of C are included in those of A except for X ;
- $A \vdash C$ is provable;

46 ■ for all formulas B such that X does not appear in B and $A \vdash B$ is provable, the sequent
 47 $C \vdash B$ is also provable.

48 In particular, uniform interpolants are an interpretation of first-order quantifiers in the
 49 logic under consideration, as the formula C above behaves like $\exists X A$ (at least for provability
 50 purposes). Not every logic has uniform interpolation, for instance the modal logic S4 does
 51 not [10] even though it has Craig's and Lyndon's interpolations [17, 9]. It is easy to check
 52 that classical logic has this property: a uniform interpolant for a formula A with respect to X
 53 is simply $A[\top/X] \vee A[\perp/X]$, that is the disjunction of substituting X by true or false. The
 54 foundational work of Pitts [18] proves that intuitionistic logic has the uniform interpolation
 55 property. The algorithm computing the interpolant is more complex than in the classical
 56 case, and the proof technique is based on a syntactic approach. This method by Pitts was
 57 then adapted to many other logics. In particular, it was used by Alizadeh, Derakhshan and
 58 Ono [1] to prove uniform interpolation for the Lambek calculus, as well as for classical and
 59 intuitionistic multiplicative-additive linear logic – and also for the affine variants of these
 60 two. This is, at the best of the author's knowledge, the sole result on uniform interpolation
 61 in linear logic to this day. Pitts' method does not extend easily to full propositional linear
 62 logic because of the contraction rule, that has a premise sequent “bigger” than its conclusion
 63 sequent, and that necessitated a particular treatment in Pitts' paper [18] so as to remove it
 64 from the calculus.

65 Nonetheless, since classical logic and intuitionistic logic both have the uniform interpola-
 66 tion property, one could expect linear logic to also have it. We show it is not the case by
 67 exhibiting a counter-example. Actually, we present formulas A and $(B_n)_{n \in \mathbb{N}}$ such that:

- 68 1. an atom X appears in A but not in any of the B_n ;
- 69 2. $A \vdash B_n$ is provable for all number n ;
- 70 3. for all formulas C such that X does not appear in C , for any number $n \in \mathbb{N}$, if $A \vdash C$
 71 and $C \vdash B_n$ are both provable, then C is of size at least n .

72 The result immediately follows: A has no uniform interpolant with respect to X , since such
 73 an interpolant would have an unbounded size. Furthermore, the counter-example we exhibit
 74 is not only one for uniform interpolation in linear logic, but also in its intuitionistic and affine
 75 variants. A subtle use of the exponential is mandatory inside the formula A , since uniform
 76 interpolation holds with no exponential (case of multiplicative-additive linear logic) and with
 77 exponentials everywhere (case of classical logic). This complexity of A entails difficulties
 78 to prove some results for sequents of the shape $A \vdash C$. In particular, the most challenging
 79 part in this paper is the proof of property 3, that involves some technical tools among which
 80 geometry of interaction.

81 **Outline.** We first recall the syntax of linear logic in Section 2, along with a definition of
 82 uniform interpolation in this setting. Then, we define some useful but technical tools in
 83 Section 3. The main part, Section 4, consists of a presentation of our counter-example, and a
 84 proof it cannot have a uniform interpolant using the tools from the preceding section. Finally,
 85 we show in Section 5 that our counter-example is also one to the uniform interpolation
 86 property of *intuitionistic* linear logic and of *affine* logic (both classical and intuitionistic).

87 2 Linear Logic

88 We use the standard presentation of linear logic as a unilateral sequent calculus [11]. In
 89 particular, a proof is a *derivation* whose conclusion sequent is of the shape $\vdash \Gamma$.

$$\begin{array}{c}
\frac{}{\vdash X^+, X^-} (ax) \quad \frac{\vdash \Gamma, A \quad \vdash A^\perp, \Delta}{\vdash \Gamma, \Delta} (cut) \\
\\
\frac{\vdash A, B, \Gamma}{\vdash A \wp B, \Gamma} (\wp) \quad \frac{\vdash A, \Gamma \quad \vdash B, \Delta}{\vdash A \otimes B, \Gamma, \Delta} (\otimes) \quad \frac{\vdash \Gamma}{\vdash \perp, \Gamma} (\perp) \quad \frac{}{\vdash 1} (1) \\
\\
\frac{\vdash A, \Gamma \quad \vdash B, \Gamma}{\vdash A \& B, \Gamma} (\&) \quad \frac{\vdash A, \Gamma}{\vdash A \oplus B, \Gamma} (\oplus_1) \quad \frac{\vdash B, \Gamma}{\vdash A \oplus B, \Gamma} (\oplus_2) \quad \frac{}{\vdash \top, \Gamma} (\top) \\
\\
\frac{\vdash A, \Gamma}{\vdash ?A, \Gamma} (?d) \quad \frac{\vdash ?A, ?A, \Gamma}{\vdash ?A, \Gamma} (?c) \quad \frac{\vdash \Gamma}{\vdash ?A, \Gamma} (?w) \quad \frac{\vdash A, ?\Gamma}{\vdash !A, ?\Gamma} (!)
\end{array}$$

■ **Figure 1** Rules of Linear Logic

90 **Formulas** are defined by the following grammar, with X an atom in some countable set:

$$91 \quad A, B ::= \underbrace{X^- \mid X^+}_{\text{atom}} \mid \underbrace{A \wp B \mid A \otimes B \mid \perp \mid 1}_{\text{multiplicative}} \mid \underbrace{A \& B \mid A \oplus B \mid \top \mid 0}_{\text{additive}} \mid \underbrace{?A \mid !A}_{\text{exponential}}$$

92 **Orthogonality** $(\cdot)^\perp$ (*a.k.a.* negation, duality) is an involution defined inductively by:

$$\begin{array}{llll}
93 \quad (X^-)^\perp \stackrel{\text{def}}{=} X^+ & (A \wp B)^\perp \stackrel{\text{def}}{=} A^\perp \otimes B^\perp & (A \& B)^\perp \stackrel{\text{def}}{=} A^\perp \oplus B^\perp & (?A)^\perp \stackrel{\text{def}}{=} !A^\perp \\
(X^+)^\perp \stackrel{\text{def}}{=} X^- & (A \otimes B)^\perp \stackrel{\text{def}}{=} A^\perp \wp B^\perp & (A \oplus B)^\perp \stackrel{\text{def}}{=} A^\perp \& B^\perp & (!A)^\perp \stackrel{\text{def}}{=} ?A^\perp \\
& \perp^\perp \stackrel{\text{def}}{=} 1 & \top^\perp \stackrel{\text{def}}{=} 0 & \\
& 1^\perp \stackrel{\text{def}}{=} \perp & 0^\perp \stackrel{\text{def}}{=} \top &
\end{array}$$

94 **Linear implication** is defined as $A \multimap B \stackrel{\text{def}}{=} A^\perp \wp B$. **Rules** are given on Figure 1, where
95 A and B stand for arbitrary formulas, Γ and Δ for sets of (occurrences of) formulas, and $?\Gamma$
96 for a set of (occurrences of) $?$ -formulas. Remark we restrict the *ax*-rule to *atomic formulas*,
97 a well-known simplification. This sequent calculus is equipped with a **cut-elimination**
98 procedure, denoted by $\longrightarrow$, that is weakly normalising and that we recall in Appendix A.

99 Uniform interpolation in linear logic can be stated as follows. We denote by $\mathcal{V}(A)$ the
100 set of atoms occurring in the formula A – e.g. $\mathcal{V}((X^+ \wp X^-) \& Y^-) = \{X, Y\}$.

101 ► **Definition 1** (Uniform Interpolant). *Given a formula A and an atom X , a **uniform***
102 ***interpolant** for A with respect to X is a formula C such that:*

- 103 ■ $\mathcal{V}(C) \subseteq \mathcal{V}(A) \setminus \{X\}$;
- 104 ■ $\vdash A^\perp, C$ is provable;
- 105 ■ for all formulas B , if $X \notin \mathcal{V}(B)$ and $\vdash A^\perp, B$ is provable then $\vdash C^\perp, B$ is provable.

106 The **uniform interpolation property** holds when every formula admits a uniform inter-
107 polant with respect to every atom.

108 3 Toolbox

109 We present here two tools used to prove that our counter-example has no uniform interpolant.
110 The first is a variant of *slices*, the second of *Geometry Of Interaction*. They are needed solely
111 to prove our most technical statement, corresponding to property 3 in the introduction.

112 3.1 A variant of Slices

113 The $\&$ -rule stands apart in linear logic since it is the sole rule with some sharing of the
114 context Γ . A useful tool when considering $\&$ -rules is the notion of a *slice* [11, 13] which is a
115 derivation missing some additive component: a slice is obtained from a derivation by deleting

one sub-tree of each $\&$ -rule. In a slice, no rule duplicates a context. As our framework has not only additive but also exponential rules, the well-behaved concept is the one of a 0-slice, meaning we slice $\&$ -rules that are not above a $!$ -rule, *i.e.* those at depth 0.

► **Definition 2** (0-slice). For π a derivation, a 0-slice of π is a (usually non-derivation) tree π' obtained by deleting one of the two sub-trees of each $\&$ -rule of π that is not above a $!$ -rule. Thus, in π' some (but maybe not all) $\&$ -rules are unary:

$$\frac{\vdash A, \Gamma}{\vdash A \& B, \Gamma} (\&_1) \quad \frac{\vdash B, \Gamma}{\vdash A \& B, \Gamma} (\&_2)$$

► **Example 3.** Here is a derivation followed by its two 0-slices:

$$\begin{array}{c} \frac{\frac{\frac{}{\vdash \top, !(1 \& 1)} (\top)}{\vdash \top, !(1 \& 1)} (\&_1) \quad \frac{\frac{\frac{}{\vdash \perp, !(1 \& 1)} (\perp)}{\vdash \perp, !(1 \& 1)} (\&_2)}{\vdash \top \& \perp, !(1 \& 1)} (\&) \\ \frac{\frac{\frac{}{\vdash \top, !(1 \& 1)} (\top)}{\vdash \top, !(1 \& 1)} (\&_1) \quad \frac{\frac{\frac{}{\vdash \perp, !(1 \& 1)} (\perp)}{\vdash \perp, !(1 \& 1)} (\&_2)}{\vdash \top \& \perp, !(1 \& 1)} (\&) \end{array}$$

Cut-elimination can be defined in the natural way for 0-slices, taking the definition for derivations and extending the $\& - \oplus$ key case with

$$\begin{array}{c} \frac{\frac{\vdash A, \Gamma}{\vdash A \& B, \Gamma} (\&_1) \quad \frac{\vdash A^\perp, \Delta}{\vdash A^\perp \oplus B^\perp, \Delta} (\oplus_1)}{\vdash \Gamma, \Delta} (cut) \longrightarrow \frac{\vdash \Gamma, A \quad \vdash A^\perp, \Delta}{\vdash \Gamma, \Delta} (cut) \\ \frac{\frac{\vdash B, \Gamma}{\vdash A \& B, \Gamma} (\&_2) \quad \frac{\vdash B^\perp, \Delta}{\vdash A^\perp \oplus B^\perp, \Delta} (\oplus_2)}{\vdash \Gamma, \Delta} (cut) \longrightarrow \frac{\vdash \Gamma, B \quad \vdash B^\perp, \Delta}{\vdash \Gamma, \Delta} (cut) \end{array}$$

as well as replacing $\&$ -rules with $\&_1$ - or $\&_2$ -rules during a $?_d - !$ key case (where a $!$ -rule is removed). Observe it is not possible to eliminate *cut*-rules of the two following shapes:

$$\frac{\frac{\vdash A, \Gamma}{\vdash A \& B, \Gamma} (\&_1) \quad \frac{\vdash B^\perp, \Delta}{\vdash A^\perp \oplus B^\perp, \Delta} (\oplus_2)}{\vdash \Gamma, \Delta} (cut) \quad \frac{\frac{\vdash B, \Gamma}{\vdash A \& B, \Gamma} (\&_2) \quad \frac{\vdash A^\perp, \Delta}{\vdash A^\perp \oplus B^\perp, \Delta} (\oplus_1)}{\vdash \Gamma, \Delta} (cut)$$

The only property of 0-slices we need is their behaviour with respect to cut-elimination.

► **Lemma 4.** Let π and ϕ be derivations such that $\pi \longrightarrow \phi$. For each 0-slice ϕ' of ϕ , there exists a 0-slice π' of π such that $\pi' \longrightarrow \phi'$ or $\pi' = \phi'$.

Proof. We can check that each cut-elimination step respects this property, with the equality case coming from a reduction on rules not in the considered 0-slice. ◀

► **Remark 5.** Observe a corresponding result for slices does not hold, because a formula of the shape $?(A \oplus B)$ can be associated to both a $\oplus_1$ - and a $\oplus_2$ -rule. This is a phenomenon at play, for instance, in the Seely isomorphism $?(A \oplus B) \simeq ?A \wp ?B$. The cut-elimination case responsible for this failure is the $?_c - !$ key case, because the two copies of the duplicated derivation (the “ $!$ box”) may not make the same choices for all of their corresponding $\&$ -rules.

3.2 A variant of Geometry Of Interaction

We define here the key tool for proving our counter-example has no uniform interpolant, previously used in the restricted setting of unit-free multiplicative-additive linear logic [6]. It falls within the spirit of *Geometry Of Interaction (GOI)* [12]. The quite unusual part is that we will look at *connectives* of the *cut*-formulas along a path from GOI.

► **Definition 6** (GOI projection graph). *The **GOI projection graph** $\mathcal{G}_\pi$ of a derivation, or of a 0-slice, π of conclusion $\vdash \Gamma$ is the undirected edge-bicoloured graph defined as follows, colors being **ax** and **cut**.*

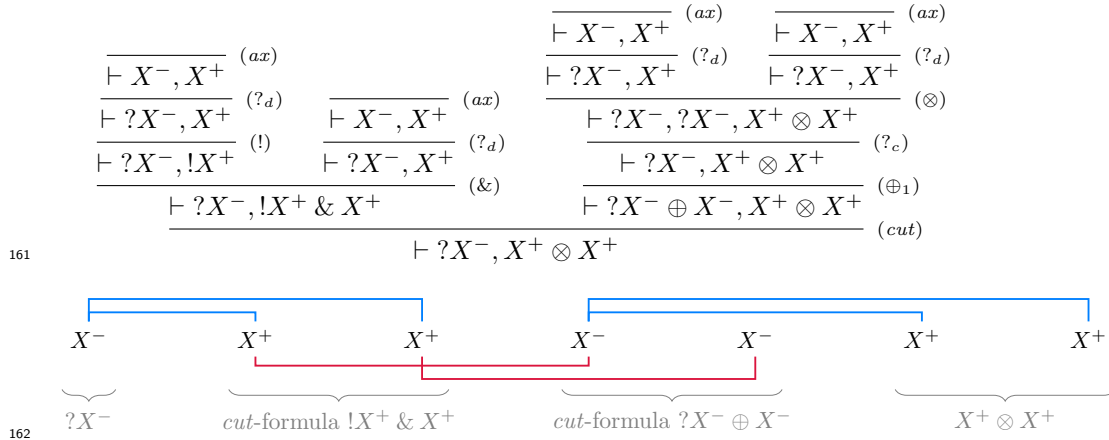
Vertices: All occurrences of atomic formulas X^+ and X^- (for all atoms) from the formulas of Γ and from the cut-formulas A and $A^\perp$ in π , with each cut-rule yielding a pair of cut-formulas giving separate and dual occurrences.

ax-edges: For each pair of occurrences X^+ and X^- such that π contains an ax-rule $\frac{}{\vdash X^+, X^-} (ax)$, there is an ax-edge between X^+ and X^- .

cut-edges: For each cut-rule $\frac{\vdash \Delta, A \quad \vdash A^\perp, \Sigma}{\vdash \Delta, \Sigma} (cut)$ in π , there is a cut-edge between dual occurrences from the two formulas A and $A^\perp$.

In our figures, **ax**-edges are coloured in **azure** and depicted above the vertices, whereas **cut**-edges are coloured in **crimson** and depicted below the vertices.

► **Example 7.** Here is a derivation π followed by its GOI projection graph $\mathcal{G}_\pi$:



► **Remark 8.** Observe they are two ways to obtain several **ax**-edges on a same occurrence: this occurrence can be duplicated using a $?_c$ -rule on one of its ancestor (in the formula tree), or by being in the context of a $\&$ -rule. Both cases can be observed in Example 7.

We will consider **alternating** paths in a GOI projection graph, which are paths¹ where consecutive edges are coloured differently – e.g. we start with an **ax**-edge, then take a **cut**-edge, then an **ax**-edge, and so on. We have the usual result of GOI about *persistent* paths, i.e. some paths are preserved through cut-elimination [3]. What is relevant for our purposes is that this holds not only for derivations but also for 0-slices.

¹ A path is a finite alternating sequence of vertices and edges $(v_0, e_1, v_1, e_2, v_2, \dots, e_n, v_n)$ such that for all $i \in \{1, \dots, n\}$, the two endpoints of e_i are v_{i-1} and v_i .

171 ► **Lemma 9.** Consider a cut-elimination step $\pi \rightarrow \phi$ between two derivations, or between
 172 two 0-slices, of conclusion $\vdash \Gamma$. Let u and v be occurrences of atoms from Γ . If there is an
 173 alternating path between u and v in $\mathcal{G}_\phi$, then there is an alternating path between them in $\mathcal{G}_\pi$.

174 **Proof.** By inspection on the kind of the cut-elimination step. Since we only care about
 175 ax -rules and atoms, the non-immediate cases are either trivial with $\mathcal{G}_\phi \subseteq \mathcal{G}_\pi$ ($\& - \oplus$ key
 176 cases, $?_d - !$ key case, $?_w - !$ key case, $\top - cut$ commutative case) or easy to check (ax key
 177 case, $?_c - !$ key case). ◀

178 ► **Remark 10.** It is easy to find a counter-example to the reciprocal of Lemma 9. One could
 179 be tempted to consider not only ax -rules but also $\top$ -rules in a GOI projection graph, also
 180 adding vertices for all occurrences of units. Such a modified setting does not suit our purposes
 181 for Lemma 9 would not hold – consider *e.g.*:

$$\begin{array}{c}
 \frac{\frac{\vdash \top, \perp \wp 1}{\vdash \top, \perp \wp 1} (\top) \quad \frac{\frac{\frac{\vdash 1}{\vdash 1} (1) \quad \frac{\frac{\vdash X^+, X^-}{\vdash X^+, X^-} (ax)}{\vdash \perp, X^+, X^-} (\perp)}{\vdash 1 \otimes \perp, X^+, X^-} (\otimes)}{\vdash \top, X^+, X^-} (cut) \rightarrow \frac{\vdash \top, X^+, X^-}{\vdash \top, X^+, X^-} (\top)
 \end{array}$$

182

183 4 A Counter-Example to Uniform Interpolation

184 Let us fix two atoms X and Y . Our counter-example is:

$$185 \quad A \stackrel{\text{def}}{=} X^+ \otimes ! (X^+ \multimap (X^+ \otimes Y^+)) \otimes (X^+ \multimap Y^+) \quad \begin{cases} B_0 & \stackrel{\text{def}}{=} Y^+ \otimes Y^+ \\ B_{n+1} & \stackrel{\text{def}}{=} B_n \otimes Y^+ \end{cases} \quad (1)$$

186 The main intuition is the following. The formula A is made of three parts: X^+ , a “reusable
 187 machine” taking as input a X^+ and returning a X^+ and a Y^+ , and a “one-use machine”
 188 taking as input a X^+ and returning a Y^+ . Thence, A can “produce” any non-null number
 189 of Y^+ , so that $A \vdash B_n$ is provable for all $n \in \mathbb{N}$. Therefore, one may think that $!Y^+ \otimes Y^+$
 190 is a uniform interpolant for A , and indeed we do have $!Y^+ \otimes Y^+ \vdash B_n$ provable; however,
 191 $A \vdash !Y^+ \otimes Y^+$ cannot be proved.

192 Let us mention that A can be viewed as some “pseudo ! exponential” of Y^+ , since the
 193 following dereliction and selection rules for Y^- are admissible:

$$194 \quad \frac{\vdash Y^-, \Gamma}{\vdash A^\perp, \Gamma} \quad \frac{\vdash Y^-, A^\perp, \Gamma}{\vdash A^\perp, \Gamma}$$

195 It is reminiscent of two other “pseudo ! exponential” of Y^+ from the literature:

- 196 ■ $Y^+ \otimes ! (Y^+ \multimap (Y^+ \otimes Y^+)) \otimes ! (Y^+ \multimap 1)$ used by Yves Lafont to discuss some non-
 197 admissible !-introduction rule [15, page 14];
- 198 ■ $X^+ \otimes ! (X^+ \multimap (X^+ \otimes X^+)) \otimes ! (X^+ \multimap 1) \otimes ! (X^+ \multimap Y^+)$ (in fact a formula isomorphic
 199 to it) used by Anupam Das to prove different the exponentials of linear logic and their
 200 encoding in multiplicative-additive linear logic with least and greatest fixed points [4].

201 The later formula by Anupam Das could also be used here in place of our A . Nonetheless,
 202 we preferred the definition of Equation (1) because it is smaller and we do not need the full
 203 power of an exponential – in particular not the $?_w$ -rule.

To obtain that there is no uniform interpolant for A , it stays to check the three properties claimed in the introduction: $X \notin \mathcal{V}(B_n)$; and $A \vdash B_n$ is provable; and for all formulas C such that $X \notin \mathcal{V}(C)$, if $A \vdash C$ and $C \vdash B_n$ are both provable, then C is of size at least n . Note that the first property holds by definition. The second is easy to prove.

► **Lemma 11.** *For all $n \in \mathbb{N}$, there is a derivation of $\vdash A^\perp, B_n$.*

Proof. We build a derivation π_n of $\vdash X^-, ?(X^+ \otimes (X^- \wp Y^-)), X^+ \otimes Y^-, B_n$ by induction on $n \in \mathbb{N}$. The result then follows by exhibiting:

$$\begin{array}{c}
 \frac{\frac{\vdash X^-, ?(X^+ \otimes (X^- \wp Y^-)), X^+ \otimes Y^-, B_n}{\vdash X^-, ?(X^+ \otimes (X^- \wp Y^-)) \wp (X^+ \otimes Y^-), B_n} (\wp)}{\vdash X^- \wp ?(X^+ \otimes (X^- \wp Y^-)) \wp (X^+ \otimes Y^-), B_n} (\wp) \\
 \pi_0 \stackrel{\text{def}}{=} \left\{ \begin{array}{l}
 \frac{\frac{\frac{\frac{\vdash X^-, X^+}{\vdash X^-, X^+} (ax) \quad \frac{\frac{\frac{\vdash Y^-, Y^+}{\vdash Y^-, Y^+} (ax) \quad \frac{\vdash Y^-, Y^+}{\vdash Y^-, Y^+} (ax)}{\vdash Y^-, Y^-, Y^+ \otimes Y^+} (\otimes)}{\vdash X^-, Y^-, X^+ \otimes Y^-, Y^+ \otimes Y^+} (\otimes)}{\vdash X^-, Y^-, ?(X^+ \otimes (X^- \wp Y^-)), X^+ \otimes Y^-, Y^+ \otimes Y^+} (\wp)}{\vdash X^-, X^+ \otimes (X^- \wp Y^-), X^+ \otimes Y^-, Y^+ \otimes Y^+} (\otimes)}{\vdash X^-, ?(X^+ \otimes (X^- \wp Y^-)), X^+ \otimes Y^-, Y^+ \otimes Y^+} (?_d) \\
 \pi_{n+1} \stackrel{\text{def}}{=} \frac{\frac{\frac{\frac{\frac{\vdash X^-, X^+}{\vdash X^-, X^+} (ax) \quad \frac{\frac{\frac{\frac{\frac{\vdash ?(X^+ \otimes (X^- \wp Y^-)), X^+ \otimes Y^-, B_n}{\vdash X^-, Y^-, ?(X^+ \otimes (X^- \wp Y^-)), X^+ \otimes Y^-, B_n \otimes Y^+} (\otimes)}{\vdash X^-, Y^-, ?(X^+ \otimes (X^- \wp Y^-)), X^+ \otimes Y^-, B_n \otimes Y^+} (\wp)}{\vdash X^-, X^+ \otimes (X^- \wp Y^-), ?(X^+ \otimes (X^- \wp Y^-)), X^+ \otimes Y^-, B_n \otimes Y^+} (\otimes)}{\vdash X^-, ?(X^+ \otimes (X^- \wp Y^-)), ?(X^+ \otimes (X^- \wp Y^-)), X^+ \otimes Y^-, B_n \otimes Y^+} (?_d)}{\vdash X^-, ?(X^+ \otimes (X^- \wp Y^-)), X^+ \otimes Y^-, B_n \otimes Y^+} (?_c)
 \end{array} \right.
 \end{array}$$

The third wished property is more complex to prove: we use the tools from Section 3. It also needs an intermediate result, that is not surprising, albeit a little knotty to prove.

► **Lemma 12.** *Let C be a formula such that $X \notin \mathcal{V}(C)$ and $\vdash A^\perp, C$ is provable. There exists a cut-free derivation of $\vdash A^\perp, C$ in which no $!$ -rule has $?(X^+ \otimes (X^- \wp Y^-))$ in its conclusion sequent.*

Proof. Consider a cut-free derivation π of $\vdash A^\perp, C$ with a minimal number of $!$ -rules having $?(X^+ \otimes (X^- \wp Y^-))$ in their conclusion sequents. We prove this number is null.

By the sub-formula property, any given $!$ -rule in π is of the shape

$$\frac{\phi}{\vdash (?(X^+ \otimes (X^- \wp Y^-)))^i, D, ?\Delta} (!)$$

with a conclusion sequent containing $i \geq 0$ copies of $?(X^+ \otimes (X^- \wp Y^-))$ for some $i \in \mathbb{N}$, and where $!D, ?\Delta$ are sub-formulas of C . Towards a contradiction, assume $i \geq 1$. Substituting

226 X by 0 in ϕ yields a derivation $\phi[0/X]$ of $\vdash (? (0 \otimes (\top \wp Y^-)))^i, D, ?\Delta$ since, by hypothesis,
 227 there is no X in $D, ?\Delta$. Consider any result τ of fully eliminating *cut*-rules in

$$\begin{array}{c}
 \frac{}{\vdash \top, 0 \otimes Y^+} (\top) \\
 \frac{}{\vdash \top \wp (0 \otimes Y^+)} (\wp) \\
 \frac{}{\vdash !(\top \wp (0 \otimes Y^+))} (!) \\
 \frac{}{\vdash D, ?\Delta} (\top) \\
 \frac{}{\vdash !D, ?\Delta} (!) \\
 \frac{}{\vdash (? (X^+ \otimes (X^- \wp Y^-)))^i, !D, ?\Delta} (?_w)
 \end{array}$$

228

229 where doubled inference rules mean we apply the rule several times depending on i . Then,
 230 τ has no $!$ -rule having $? (X^+ \otimes (X^- \wp Y^-))$ in its conclusion sequent by the sub-formula
 231 property. Therefore, the derivation obtained from π by substituting with τ the sub-tree

$$\frac{}{\vdash (? (X^+ \otimes (X^- \wp Y^-)))^i, D, ?\Delta} (!)$$

232 $\vdash (? (X^+ \otimes (X^- \wp Y^-)))^i, !D, ?\Delta$ contradicts the minimality assumption on π . $\blacktriangleleft$

233 **► Proposition 13.** *Let C be a formula such that $X \notin \mathcal{V}(C)$. If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are*
 234 *both provable for some number $n \in \mathbb{N}$, then C contains at least $n + 2$ occurrences of Y^+ .*

235 **Proof.** Let us call π and ϕ derivations of respectively $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$. We assume
 236 them *cut*-free without any loss of generality. We will use the GOI projection graph of (some
 237 0-slice of) their composition by *cut* to identify Y^- 's in $C^\perp$ linked to the Y^+ 's of B_n . Thanks
 238 to alternating paths, we will prove that these occurrences of Y^- are pairwise distinct. This
 239 yields one Y^- in $C^\perp$ for each Y^+ in B_n , concluding the proof.

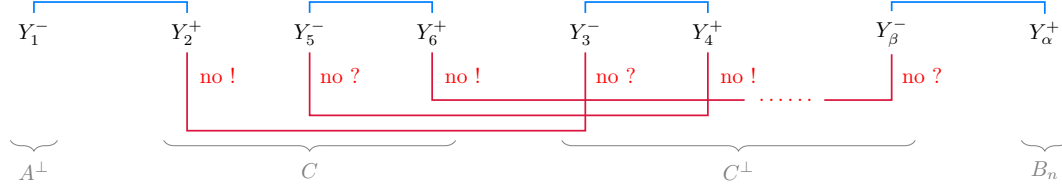
240 **Finding an alternating path in some 0-slice.** Observe that $\frac{\pi}{\vdash A^\perp, C} \quad \frac{\phi}{\vdash C^\perp, B_n} \xrightarrow{(cut)} \vdash A^\perp, B_n$

241 reduces to a *cut*-free derivation of $\vdash A^\perp, B_n$, that we call τ . This τ is its own unique 0-slice,
 242 since there is no $\&$ -connective in $A^\perp$ nor in B_n . By repeated applications of Lemma 4, there

243 is a 0-slice μ of $\frac{\pi}{\vdash A^\perp, C} \quad \frac{\phi}{\vdash C^\perp, B_n} \xrightarrow{(cut)} \vdash A^\perp, B_n$ such that $\mu \rightarrow^* \tau$.

244 Consider some occurrence Y_α^+ in B_n , with the index α used to distinguish it from other
 245 occurrences. Since τ is a *cut*-free derivation of $\vdash A^\perp, B_n$, there must be an *ax*-rule between
 246 this occurrence Y_α^+ and one of the two occurrences of Y^- in $A^\perp$: call it Y_1^- . Hence, there is an
 247 alternating path (made of one *ax*-edge) between these two occurrences in the GOI projection
 248 graph $\mathcal{G}_\tau$ of τ . By repeated applications of Lemma 9, there also is an alternating path p
 249 between Y_1^- and Y_α^+ in $\mathcal{G}_\mu$. Thus, there is an occurrence Y_β^- and an *ax*-rule $\frac{}{\vdash Y_\beta^-, Y_\alpha^+} (ax)$
 250 and $Y_\beta^- - Y_\alpha^+$ is the last edge of p . This occurrence Y_β^- can only be in $C^\perp$.

251 We use the alternating path p to show there is no sub-formula $?D$ of $C^\perp$ containing Y_β^- –
 252 say differently, there is no $?$ between the root of the formula $C^\perp$ and Y_β^- . Once this claim is
 253 proved, we are done: each occurrence of Y^+ in B_n is linked by an *ax*-edge to an occurrence
 254 of Y^- in $C^\perp$. All these occurrences of Y^- are pairwise distinct (recall Remark 8): there
 255 is no sub-formula $?D$ of $C^\perp$ containing such an occurrence, so they cannot be contracted;



■ **Figure 2** Alternating path in the GOI projection graph $\mathcal{G}_\mu$ in the proof of Proposition 13

and there is no $\&$ -rule that can share the identified ax -rules since we are in a 0-slice, and these ax -rules cannot be above a $!$ -rule for there is no $?$ -connective nor $!$ -connective in B_n . Therefore, we have found $n + 2$ occurrences of Y^- in $C^\perp$, as wished.

Exploiting the alternating path. We prove our claim by showing a stronger result:

- for all occurrences of Y^+ in C (respectively in $C^\perp$) belonging to p , there is no $!$ between the root of C (respectively of $C^\perp$) and these occurrences;
- for all occurrences of Y^- in C (respectively in $C^\perp$) belonging to p , there is no $?$ between the root of C (respectively of $C^\perp$) and these occurrences.

We proceed by following the edges of p starting from its endpoint Y_1^- in $A^\perp$.

Exploiting the alternating path: base case. Consider the first edge $Y_1^- - Y_2^+$ of p , with Y_2^+ in C . Using Lemma 12, without any loss of generality, there is no $!$ -connective between the root of C and Y_2^+ . Otherwise, since there is an ax -rule on $\vdash Y_1^-, Y_2^+$, there would be a $!$ -rule (associated to this $!$ -connective) with in its conclusion sequent a $?$ -sub-formula of $A^\perp$ containing Y_1^- , meaning with $?(X^+ \otimes (X^- \wp Y^-))$: contradiction.

Exploiting the alternating path: inductive case. See Figure 2 for an illustration of our reasoning. Assume the results holds for a prefix of p , and let us consider its last edge.

1. Suppose this prefix ends with an ax -edge $Y_k^- - Y_{k+1}^+$ with Y_{k+1}^+ in C . We know there is no $!$ between the root of C and Y_{k+1}^+ . As p is alternating, the subsequent edge is a cut -edge $Y_{k+1}^+ - Y_{k+2}^-$ with Y_{k+2}^- in $C^\perp$. Since Y_{k+1}^+ and Y_{k+2}^- are dual occurrences by definition of a cut -edge, there is no $?$ between the root of $C^\perp$ and Y_{k+2}^- .
2. Suppose this prefix ends with a cut -edge $Y_k^+ - Y_{k+1}^-$ with Y_{k+1}^- in $C^\perp$. We know there is no $?$ between the root of $C^\perp$ and Y_{k+1}^- . As p is alternating, the subsequent edge is an ax -edge $Y_{k+1}^- - Y_{k+2}^+$ with Y_{k+2}^+ either in B_n or in $C^\perp$. In the first case, we are done: we have $Y_{k+2}^+ = Y_\alpha^+$ since there is no cut -edge with endpoint Y_{k+2}^+ to continue our path, and p is supposed to end in Y_α^+ . In the second case, since there is an ax -rule $\frac{}{\vdash Y_{k+1}^-, Y_{k+2}^+} (ax)$ by definition of an ax -edge, it follows there is no $!$ between the root of $C^\perp$ and Y_{k+2}^+ – because the context condition of a corresponding $!$ -rule would prevent the presence of Y_{k+1}^- in all sequents above it.
3. Suppose this prefix ends with an ax -edge $Y_k^- - Y_{k+1}^+$ with Y_{k+1}^+ in $C^\perp$. This case is identical to case 1 up to switching C and $C^\perp$.
4. Suppose this prefix ends with a cut -edge $Y_k^+ - Y_{k+1}^-$ with Y_{k+1}^- in C . This case is identical to case 2, with no need to consider Y_{k+2}^+ in $A^\perp$ since $A^\perp$ has no Y^+ .

This proves our claim, and concludes the proof. ◀

► **Remark 14.** By going in the other direction of the path p , we can similarly show that:

- 290 ■ for all occurrences of Y^+ in C (respectively in $C^\perp$) belonging to p , there is no $?$ between
 291 the root of C (respectively of $C^\perp$) and these occurrences;
- 292 ■ for all occurrences of Y^- in C (respectively in $C^\perp$) belonging to p , there is no $!$ between
 293 the root of C (respectively of $C^\perp$) and these occurrences.

294 This means the “useful” part of C as identified by p is exponential-free.

295 Now that we have all needed properties, we easily obtain our main result.

296 ► **Theorem 15.** *Linear logic does not have the uniform interpolation property.*

297 **Proof.** The formula A cannot have a uniform interpolant C with respect to X , as such a C
 298 would have an unbounded number of occurrences of Y^+ by Lemma 11 and Proposition 13. ◀

299 5 Adaptation to the Intuitionistic and Affine variants

300 Our counter-example to the uniform interpolation property for linear logic is also a counter-
 301 example for *intuitionistic* linear logic [14]. This can be directly obtained through a result of
 302 Harold Schellinx [21]. Formulas of intuitionistic linear logic are given by:

$$303 \quad A, B ::= X^+ \mid A \multimap B \mid A \otimes B \mid 1 \mid A \& B \mid A \oplus B \mid \top \mid 0 \mid !A$$

304 ► **Proposition 16** ([21, Proposition 3.8]). *Consider a formula D of intuitionist linear logic
 305 which does not contain 0 or does not contain $\multimap$. Then $\vdash D$ is provable in linear logic if and
 306 only if it is provable in intuitionistic linear logic.*

307 ► **Theorem 17.** *Intuitionistic linear logic does not have the uniform interpolation property.*

308 **Proof.** The formula A from Equation (1) has no uniform interpolant with respect to X . In-
 309 deed, $\vdash A \multimap B_n$ is provable intuitionistically for all $n \in \mathbb{N}$ using Lemma 11 and Proposition 16
 310 – for 0 is not in $A \multimap B_n$. This yields an intuitionistic derivation of $A \vdash B_n$:

$$311 \quad \frac{\frac{\overline{A \vdash A} \quad (\text{ax}) \quad \frac{\overline{B_n \vdash B_n} \quad (\text{ax})}{A, A \multimap B_n \vdash B_n} \quad (\multimap)}{\vdash A \multimap B_n} \quad (\text{cut})}{A \vdash B_n}$$

312 Hence, we conclude as in the classical case: an interpolant for A would have an unbounded
 313 number of occurrences of Y^+ by Proposition 13 and conservativity – *i.e.* a sequent provable
 314 in intuitionistic linear logic is *a fortiori* provable in linear logic. ◀

315 Our counter-example is also one to the uniform interpolation property for *affine* linear
 316 logic, both classical and intuitionistic. Affine logic is defined as linear logic where the
 317 $?w$ -rule is replaced by the more general $\frac{\vdash \Gamma}{\vdash A, \Gamma} \quad (w)$. It is equipped with a weakly normalising
 318 cut-elimination procedure – see Appendix B. We need here one more intermediate result.

319 ► **Lemma 18.** *In affine logic, there is no derivation of $\Gamma \vdash \perp$ where Γ is a sequent whose
 320 formulas are included in $\{X, !(X \multimap (X \otimes Y)), X \multimap (X \otimes Y), X \otimes Y, Y, X \multimap Y\}$, with a
 321 same formula appearing possibly several times in Γ .*

322 **Proof.** By contradiction, consider a smallest cut-free derivation of $\Gamma \vdash \perp$ with Γ a sequent
 323 of the aforementioned shape. Its last rule is either a $\otimes$ - $\wp$ - $?d$ - $?c$ - or w -rule. In each case,
 324 one of the premises of this rule is a sequent of the aforementioned shape, which contradicts
 325 the minimality hypothesis. ◀

► **Theorem 19.** *Affine logic and intuitionistic affine logic do not have the uniform interpolation property.*

Proof. We proceed almost exactly as for linear logic, the sole difference being in the beginning of the proof of Proposition 13. There, that in the normalised derivation τ of $\vdash A^\perp, B_n$ (or $A \vdash B_n$) there is an *ax*-rule on each Y^+ of B_n is no longer immediate: we invoke Lemma 18, which implies no sub-formula of B_n can be weakened in τ – this lemma holding in affine logic, so *a fortiori* in intuitionistic affine logic by conservativity. ◀

6 Conclusion

We presented a formula of unit-free multiplicative-exponential linear logic that has no uniform interpolant in linear logic, both classical and intuitionistic, as well as in classical and intuitionistic affine logic. This contrasts sharply with the situation of classical logic and intuitionistic logic and multiplicative-additive linear logic, that all have the uniform interpolation property. Our counter-example sheds some light on when the contraction rule prevents uniform interpolation: it may do so when the formula A to interpolate contains both a non-contractible sub-formula (here X^+) and a contractible sub-formula (here $!(X^+ \multimap (X^+ \otimes Y^+))$), which is impossible in the three aforementioned logics but possible in linear logic and affine logic! Furthermore, please observe that our demonstration also holds in a variant of linear logic with no $?w$ -rule and/or with the $?c$ -rule replaced by the weaker selection rule $\frac{\vdash A, ?A, \Gamma}{\vdash ?A, \Gamma}$. This makes quite clear that what prevents uniform interpolation in linear logic is a contraction rule available only on some formulas.

The formulas A and B_n we exhibited in Equation (1) are not complex, but proving no formula C can be a uniform interpolant was quite tedious as it depends heavily on the necessary structure of C . Geometry of interaction and slices were very useful tools to this end – and we do not do how to prove Proposition 13 without using a GOI projection graph.

A natural question following the failure of the uniform interpolation property is the following decision problem: given a formula A , does it have a uniform interpolant? This is an open problem for formulas of (multiplicative-exponential) (intuitionistic) linear logic. Considering the uniform interpolation property in linear logic and its fragments, since it trivially holds in the presence of quantifiers (setting $C = \exists X A$ as an interpolant for A with respect to X), and it has been proved for multiplicative-additive linear logic [1] and now been disproved for formulas of multiplicative-exponential linear logic (even allowing the interpolant to use connectives from propositional linear logic), the principal fragment where the question stays open is the (unusual) additive-exponential linear logic. Since in this logic the contraction rule is of limited interest due to the lack of multiplicative connectives, we conjecture the method of Pitts [18] could be adapted to prove uniform interpolation in this fragment, as well as in its intuitionistic and affine variants.

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429

A Cut-Elimination in Linear Logic

430

Cut-elimination in linear logic is defined on Tables 1 and 2. It is weakly normalising [11].

ax	$\frac{\frac{\overline{\vdash A, A^\perp}^{(ax)} \quad \frac{\pi}{\vdash A, \Gamma}}{\vdash A, \Gamma}^{(cut)} \longrightarrow \frac{\pi}{\vdash A, \Gamma}$
$\wp - \otimes - 1$	$\frac{\frac{\frac{\pi}{\vdash A^\perp, B^\perp, \Gamma} \quad \frac{\phi}{\vdash A^\perp \wp B^\perp, \Gamma}}{\vdash \Gamma, \Delta, \Sigma}^{(\wp)} \quad \frac{\frac{\phi}{\vdash A, \Delta} \quad \frac{\tau}{\vdash B, \Sigma}}{\vdash A \otimes B, \Delta, \Sigma}^{(\otimes)} \longrightarrow \frac{\frac{\pi}{\vdash A^\perp, B^\perp, \Gamma} \quad \frac{\tau}{\vdash B, \Sigma}}{\vdash A^\perp, \Gamma, \Sigma}^{(cut)} \quad \frac{\phi}{\vdash A, \Delta}^{(cut)} \longrightarrow \frac{}{\vdash \Gamma, \Delta, \Sigma}$
$\wp - \otimes - 2$	$\frac{\frac{\pi}{\vdash A^\perp, B^\perp, \Gamma} \quad \frac{\phi}{\vdash A^\perp \wp B^\perp, \Gamma}}{\vdash \Gamma, \Delta, \Sigma}^{(\wp)} \quad \frac{\frac{\phi}{\vdash A, \Delta} \quad \frac{\tau}{\vdash B, \Sigma}}{\vdash A \otimes B, \Delta, \Sigma}^{(\otimes)} \longrightarrow \frac{\frac{\pi}{\vdash A^\perp, B^\perp, \Gamma} \quad \frac{\phi}{\vdash A, \Delta}}{\vdash B^\perp, \Gamma, \Sigma}^{(cut)} \quad \frac{\tau}{\vdash B, \Sigma}^{(cut)} \longrightarrow \frac{}{\vdash \Gamma, \Delta, \Sigma}$
$\perp - 1$	$\frac{\frac{\pi}{\vdash \Gamma} \quad \frac{\vdash \perp, \Gamma}{}^{(\perp)} \quad \frac{}{\vdash 1}^{(1)}}{\vdash \Gamma}^{(cut)} \longrightarrow \frac{\pi}{\vdash \Gamma}$
$\& - \oplus_1$	$\frac{\frac{\pi}{\vdash A^\perp, \Gamma} \quad \frac{\phi}{\vdash B^\perp, \Gamma}}{\vdash A^\perp \& B^\perp, \Gamma}^{(\&)} \quad \frac{\tau}{\vdash A \oplus B, \Delta}^{(\oplus_1)} \longrightarrow \frac{\frac{\phi}{\vdash A^\perp, \Gamma} \quad \frac{\tau}{\vdash A, \Delta}}{\vdash \Gamma, \Delta}^{(cut)}$
$\& - \oplus_2$	$\frac{\frac{\pi}{\vdash A^\perp, \Gamma} \quad \frac{\phi}{\vdash B^\perp, \Gamma}}{\vdash A^\perp \& B^\perp, \Gamma}^{(\&)} \quad \frac{\tau}{\vdash A \oplus B, \Delta}^{(\oplus_2)} \longrightarrow \frac{\frac{\pi}{\vdash B^\perp, \Gamma} \quad \frac{\tau}{\vdash B, \Delta}}{\vdash \Gamma, \Delta}^{(cut)}$
$?d - !$	$\frac{\frac{\pi}{\vdash A^\perp, \Gamma} \quad \frac{\phi}{\vdash A, ?\Delta}}{\vdash ?A^\perp, \Gamma}^{(?d)} \quad \frac{}{\vdash !A, ?\Delta}^{(!)} \longrightarrow \frac{\pi}{\vdash A^\perp, \Gamma} \quad \frac{\phi}{\vdash A, ?\Delta}^{(cut)} \longrightarrow \frac{}{\vdash \Gamma, ?\Delta}$
$?c - !$	$\frac{\frac{\pi}{\vdash ?A^\perp, ?A^\perp, \Gamma} \quad \frac{\phi}{\vdash A, ?\Delta}}{\vdash ?A^\perp, \Gamma}^{(?c)} \quad \frac{}{\vdash !A, ?\Delta}^{(!)} \longrightarrow \frac{\frac{\pi}{\vdash ?A^\perp, ?A^\perp, \Gamma} \quad \frac{\phi}{\vdash A, ?\Delta}}{\vdash ?A^\perp, \Gamma, ?\Delta}^{(cut)} \quad \frac{\phi}{\vdash A, ?\Delta}^{(!)} \longrightarrow \frac{}{\vdash \Gamma, ?\Delta, ?\Delta}^{(?c)} \longrightarrow \frac{}{\vdash \Gamma, ?\Delta}$
$?w - !$	$\frac{\frac{\pi}{\vdash \Gamma} \quad \frac{\phi}{\vdash A, ?\Delta}}{\vdash ?A^\perp, \Gamma}^{(?w)} \quad \frac{}{\vdash !A, ?\Delta}^{(!)} \longrightarrow \frac{\pi}{\vdash \Gamma} \quad \frac{\phi}{\vdash A, ?\Delta}^{(cut)} \longrightarrow \frac{}{\vdash \Gamma, ?\Delta}$

Table 1 Cut-elimination in Linear Logic – Key cases

$cut - cut$	$\frac{\frac{\frac{\pi}{\vdash A^\perp, B^\perp, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (cut)}{\vdash B^\perp, \Gamma, \Delta} \quad \frac{\tau}{\vdash B, \Sigma} \quad (cut)}{\vdash \Gamma, \Delta, \Sigma} \longrightarrow \frac{\frac{\frac{\pi}{\vdash A^\perp, B^\perp, \Gamma} \quad \frac{\tau}{\vdash B, \Sigma} \quad (cut)}{\vdash A^\perp, \Gamma, \Sigma} \quad \frac{\phi}{\vdash A, \Delta} \quad (cut)}{\vdash \Gamma, \Delta, \Sigma}$
$\wp - cut$	$\frac{\frac{\frac{\pi}{\vdash A^\perp, B, C, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (\wp)}{\vdash A^\perp, B \wp C, \Gamma} \quad \frac{\tau}{\vdash A, \Sigma} \quad (cut)}{\vdash B \wp C, \Gamma, \Delta} \longrightarrow \frac{\frac{\frac{\pi}{\vdash A^\perp, B, C, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (cut)}{\vdash B, C, \Gamma, \Delta} \quad \frac{\tau}{\vdash A, \Sigma} \quad (\wp)}{\vdash B \wp C, \Gamma, \Delta}$
$\otimes - cut - 1$	$\frac{\frac{\frac{\pi}{\vdash A^\perp, B, \Gamma} \quad \frac{\phi}{\vdash C, \Delta} \quad (\otimes)}{\vdash A^\perp, B \otimes C, \Gamma, \Delta} \quad \frac{\tau}{\vdash A, \Sigma} \quad (cut)}{\vdash B \otimes C, \Gamma, \Delta, \Sigma} \longrightarrow \frac{\frac{\frac{\pi}{\vdash A^\perp, B, \Gamma} \quad \frac{\tau}{\vdash A, \Sigma} \quad (cut)}{\vdash B, \Gamma, \Sigma} \quad \frac{\phi}{\vdash C, \Delta} \quad (\otimes)}{\vdash B \otimes C, \Gamma, \Delta, \Sigma}$
$\otimes - cut - 2$	$\frac{\frac{\frac{\pi}{\vdash B, \Gamma} \quad \frac{\phi}{\vdash A^\perp, C, \Delta} \quad (\otimes)}{\vdash A^\perp, B \otimes C, \Gamma, \Delta} \quad \frac{\tau}{\vdash A, \Sigma} \quad (cut)}{\vdash B \otimes C, \Gamma, \Delta, \Sigma} \longrightarrow \frac{\frac{\frac{\pi}{\vdash B, \Gamma} \quad \frac{\frac{\phi}{\vdash A^\perp, C, \Delta} \quad \frac{\tau}{\vdash A, \Sigma} \quad (cut)}{\vdash C, \Delta, \Sigma} \quad \frac{\tau}{\vdash A, \Sigma} \quad (\otimes)}{\vdash B \otimes C, \Gamma, \Delta, \Sigma}$
$\perp - cut$	$\frac{\frac{\frac{\pi}{\vdash A^\perp, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (\perp)}{\vdash A^\perp, \perp, \Gamma} \quad \frac{\tau}{\vdash A, \Sigma} \quad (cut)}{\vdash \perp, \Gamma, \Delta} \longrightarrow \frac{\frac{\frac{\pi}{\vdash A^\perp, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (cut)}{\vdash \Gamma, \Delta} \quad \frac{\tau}{\vdash A, \Sigma} \quad (\perp)}{\vdash \perp, \Gamma, \Delta}$
$\& - cut$	$\frac{\frac{\frac{\pi}{\vdash A^\perp, B, \Gamma} \quad \frac{\phi}{\vdash A^\perp, C, \Gamma} \quad (\&)}{\vdash A^\perp, B \& C, \Gamma} \quad \frac{\tau}{\vdash A, \Delta} \quad (cut)}{\vdash B \& C, \Gamma, \Delta} \longrightarrow \frac{\frac{\frac{\pi}{\vdash A^\perp, B, \Gamma} \quad \frac{\tau}{\vdash A, \Delta} \quad (cut)}{\vdash B, \Gamma, \Delta} \quad \frac{\frac{\phi}{\vdash A^\perp, C, \Gamma} \quad \frac{\tau}{\vdash A, \Delta} \quad (cut)}{\vdash C, \Gamma, \Delta} \quad (\&)}{\vdash B \& C, \Gamma, \Delta}$
$\oplus_1 - cut$	$\frac{\frac{\frac{\pi}{\vdash A^\perp, B, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (\oplus_1)}{\vdash A^\perp, B \oplus C, \Gamma} \quad \frac{\tau}{\vdash A, \Sigma} \quad (cut)}{\vdash B \oplus C, \Gamma, \Delta} \longrightarrow \frac{\frac{\frac{\pi}{\vdash A^\perp, B, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (cut)}{\vdash B, \Gamma, \Delta} \quad \frac{\tau}{\vdash A, \Sigma} \quad (\oplus_1)}{\vdash B \oplus C, \Gamma, \Delta}$
$\oplus_2 - cut$	$\frac{\frac{\frac{\pi}{\vdash A^\perp, C, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (\oplus_2)}{\vdash A^\perp, B \oplus C, \Gamma} \quad \frac{\tau}{\vdash A, \Sigma} \quad (cut)}{\vdash B \oplus C, \Gamma, \Delta} \longrightarrow \frac{\frac{\frac{\pi}{\vdash A^\perp, C, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (cut)}{\vdash C, \Gamma, \Delta} \quad \frac{\tau}{\vdash A, \Sigma} \quad (\oplus_2)}{\vdash B \oplus C, \Gamma, \Delta}$
$\top - cut$	$\frac{\frac{\frac{\pi}{\vdash A^\perp, \top, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (\top)}{\vdash A^\perp, \top, \Gamma} \quad \frac{\tau}{\vdash A, \Sigma} \quad (cut)}{\vdash \top, \Gamma, \Delta} \longrightarrow \frac{\frac{\phi}{\vdash A, \Delta} \quad (\top)}{\vdash \top, \Gamma, \Delta}$
$?d - cut$	$\frac{\frac{\frac{\pi}{\vdash A^\perp, B, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (?d)}{\vdash A^\perp, ?B, \Gamma} \quad \frac{\tau}{\vdash A, \Sigma} \quad (cut)}{\vdash ?B, \Gamma, \Delta} \longrightarrow \frac{\frac{\frac{\pi}{\vdash A^\perp, B, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (cut)}{\vdash B, \Gamma, \Delta} \quad \frac{\tau}{\vdash A, \Sigma} \quad (?d)}{\vdash ?B, \Gamma, \Delta}$
$?c - cut$	$\frac{\frac{\frac{\pi}{\vdash A^\perp, ?B, ?B, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (?c)}{\vdash A^\perp, ?B, \Gamma} \quad \frac{\tau}{\vdash A, \Sigma} \quad (cut)}{\vdash ?B, \Gamma, \Delta} \longrightarrow \frac{\frac{\frac{\pi}{\vdash A^\perp, ?B, ?B, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (cut)}{\vdash ?B, ?B, \Gamma, \Delta} \quad \frac{\tau}{\vdash A, \Sigma} \quad (?c)}{\vdash ?B, \Gamma, \Delta}$
$?w - cut$	$\frac{\frac{\frac{\pi}{\vdash A^\perp, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (?w)}{\vdash A^\perp, ?B, \Gamma} \quad \frac{\tau}{\vdash A, \Sigma} \quad (cut)}{\vdash ?B, \Gamma, \Delta} \longrightarrow \frac{\frac{\frac{\pi}{\vdash A^\perp, \Gamma} \quad \frac{\phi}{\vdash A, \Delta} \quad (cut)}{\vdash \Gamma, \Delta} \quad \frac{\tau}{\vdash A, \Sigma} \quad (?w)}{\vdash ?B, \Gamma, \Delta}$
$! - cut$	$\frac{\frac{\frac{\pi}{\vdash ?A^\perp, B, ?\Gamma} \quad \frac{\phi}{\vdash A, ?\Delta} \quad (!)}{\vdash ?A^\perp, !B, ?\Gamma} \quad \frac{\tau}{\vdash A, \Sigma} \quad (cut)}{\vdash !B, ?\Gamma, ?\Delta} \longrightarrow \frac{\frac{\frac{\pi}{\vdash ?A^\perp, B, ?\Gamma} \quad \frac{\phi}{\vdash A, ?\Delta} \quad (!)}{\vdash ?A^\perp, B, ?\Gamma} \quad \frac{\tau}{\vdash A, \Sigma} \quad (!)}{\vdash !B, ?\Gamma, ?\Delta}$

■ **Table 2** Cut-elimination in Linear Logic – Commutative cases

431 **B Cut-Elimination in Affine Logic**

432 Cut-elimination in affine logic is defined as in linear logic, except the $?w - !$ key case and the
 433 $?w - cut$ commutative case are replaced by the cases on Table 3. That cut-elimination is weakly
 434 normalising in affine logic can be proved similarly to weak normalisation of cut-elimination
 435 in linear logic – see *e.g.* the proof in [16].

w	$\frac{\frac{\pi}{\vdash \Gamma} \quad (w) \quad \vdash A^\perp, \Delta}{\vdash \Gamma, \Delta} \quad (cut) \quad \longrightarrow \quad \frac{\pi}{\vdash \Gamma} \quad (w)$
$w - cut$	$\frac{\frac{\pi}{\vdash A, \Gamma} \quad (w) \quad \frac{\phi}{\vdash A^\perp, \Delta}}{\vdash \Gamma, B, \Delta} \quad (cut) \quad \longrightarrow \quad \frac{\frac{\pi}{\vdash A, \Gamma} \quad \frac{\phi}{\vdash A^\perp, \Delta}}{\vdash \Gamma, \Delta} \quad (cut) \quad \longrightarrow \quad \frac{\pi}{\vdash \Gamma, B, \Delta} \quad (w)$

■ **Table 3** Cut-elimination in Affine Logic